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IMAGE SENSING STRUCTURE AND MANUFACTURING METHOD THEREOF

Abstract

The present disclosure provides an image sensing structure. The image sensing structure includes: a first pixel and a second pixel. The first pixel includes a first photodiode, a second photodiode, and a first isolation member. The first isolation member is disposed between the first photodiode and the second photodiode and extends along a first direction. The second pixel is disposed adjacent to the first pixel and includes a third photodiode, a fourth photodiode, and a second isolation member. The second isolation member is disposed between the third photodiode and the fourth photodiode and extends along a second direction substantially perpendicular to the first direction.

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Background/Summary

REFERENCE TO RELATED APPLICATION [0001] This Application claims the benefit of U.S. Provisional Application number 63/558, 144, filed Feb. 27, 2024, the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND

[0002] CMOS image sensors are used in many types of electronic devices, such as video cameras and digital cameras to, capture images.

[0003] As technological standards advance, there is an ever-increasing consumer demand for image-sensing devices that occupy less space, consume less power, and produce higher-quality images at greater speeds. As a result, there remains a need to develop a CMOS image sensor with an improved structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 is a schematic top view of a portion of an image sensing structure, in accordance with some embodiments of the present disclosure.

[0006] FIG. 2 is a schematic cross-sectional view of a portion of the image sensing structure along a line AA' in FIG. 1, in accordance with some embodiments of the present disclosure.

[0007] FIG. 3 is a schematic cross-sectional view of a portion of the image sensing structure along a line BB' in FIG. 1, in accordance with some embodiments of the present disclosure.

[0008] FIG. 4 is a schematic top view showing a color filter array (CFA) of another image sensing structure, in accordance with some embodiments of the present disclosure.

[0009] FIGS. 5A and 5B are enlarged top views of portions R10 and R20 of FIG. 4, respectively, in accordance with some embodiments of the present disclosure.

[0010] FIG. 6A is a schematic cross-sectional view of the portion R10 along a line CC' in FIG. 4, in accordance with some embodiments of the present disclosure.

[0011] FIG. 6B is a schematic perspective view of FIG. 6A, in accordance with some embodiments of the present disclosure.

[0012] FIG. 7A is a schematic cross-sectional view of the portion R20 along a line DD' in FIG. 4, in accordance with some embodiments of the present disclosure.

[0013] FIG. 7B is a schematic perspective view of FIG. 7A, in accordance with some embodiments of the present disclosure.

[0014] FIGS. 8 to 10 are schematic perspective views of various portions R30, R40 and R50 of FIG. 4, in accordance with some embodiments of the present disclosure.

[0015] FIG. 11 is a schematic top view showing a CFA of another image sensing structure, in accordance with some embodiments of the present disclosure.

[0016] FIG. 12A is a schematic cross-sectional view of a portion R60 along a line EE' in FIG. 11, in accordance with some embodiments of the present disclosure.

[0017] FIG. **12B** is a schematic perspective view of FIG. **12A**, in accordance with some embodiments of the present disclosure.

[0018] FIG. **13A** is a schematic cross-sectional view of a portion **R70** along a line **FF'** in FIG. **11**, in accordance with some embodiments of the present disclosure.

[0019] FIG. **13B** is a schematic perspective view of FIG. **13A**, in accordance with some embodiments of the present disclosure.

[0020] FIGS. **14** and **15** are schematic perspective views of portions **R80** and **R90** of FIG. **11**, respectively in accordance with some embodiments of the present disclosure.

[0021] FIG. **16** is a schematic top view showing a CFA of another image sensing structure, in accordance with some embodiments of the present disclosure.

[0022] FIG. **17** is a schematic top view showing a CFA of another image sensing structure, in accordance with some embodiments of the present disclosure.

[0023] FIG. **18A** is an enlarged view of a portion **R120** of the image sensing structure of FIG. **17**, in accordance with some embodiments of the present disclosure.

[0024] FIGS. **18B** and **18C** are schematic cross-sectional and perspective views of FIG. **18A**, in accordance with some embodiments of the present disclosure.

[0025] FIGS. **19A** and **19B** are schematic cross-sectional and perspective views of another portion **R130** of the image sensing structure of FIG. **17**, in accordance with some embodiments of the present disclosure.

[0026] FIG. **20** is a flow diagram showing a method for forming an image sensing structure, in accordance with some embodiments of the present disclosure.

[0027] FIGS. **21** to **27** are schematic cross-sectional or top views illustrating sequential operations of the method in FIG. **20**, in accordance with some embodiments of the present disclosure.

[0028] FIG. **28** is a schematic perspective view illustrating a combination of two image sensing structures, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0029] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In some embodiments, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0030] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass orientations of the device in use or operation in some embodiments different from the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0031] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the normal deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately” and “about” generally mean within a value or range which can be contemplated by people having ordinary skill in the art. Alternatively,

the terms “substantially,” “approximately” and “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of time, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

[0032] A CMOS image sensor can be designed as having multiple dual photodiodes (DPDs) arranged as an array. The dual photodiode can absorb more light than a single photodiode of having the same number of pixels. Therefore, the dual photodiode can provide a more efficient mechanism to generate photo-induced charges. The CMOS image sensor with DPDs can reduce noise and capture images with vibrant colors.

[0033] FIG. 1 is a schematic top view of a portion of an image sensing structure **100**, and FIG. 2 is a schematic cross-sectional view of a portion of the image sensing structure **100** along a line AA' in FIG. 1. In some embodiments, the image sensing structure **100** is an image sensing device or a portion of the image sensing device. The image sensing structure **100** is, for example, a backside illumination (BSI) image sensing structure. FIG. 1 illustrates the portion of the image sensing structure **100** in a color filter array (CFA) **127A**. In some embodiments, the image sensing structure **100** includes an optical portion **P1**, a pixel portion **P2** and a circuitry portion **P3**. The pixel portion **P2** is between the optical portion **P1** and the circuitry portion **P3**. The pixel portion **P2** includes a substrate **10** having a first surface **S1** and a second surface **S2** opposite to the first surface **S1**. The substrate **10** includes any type of semiconductor body such as a silicon wafer or one or more dies on a wafer, as well as any other type of semiconductor and/or epitaxial layers formed thereon and/or otherwise associated therewith. In some embodiments, the substrate **10** is doped with p-type impurities and thus forms a p-type substrate.

[0034] Several isolation structures **21** are disposed within the substrate **10**. In some embodiments, the isolation structure **21** is a deep trench isolation (DTI) structure. In some embodiments, the isolation structure **21** extends along a first direction **D1** between the first surface **S1** and the second surface **S2** of the substrate **10**. In some embodiments, the isolation structure **21** extends from the second surface **S2** toward a predetermined depth of the substrate **10**. In some embodiments, the first direction **D1** is a direction along a thickness of the substrate **10** or a height of the image sensing structure **100**. The isolation structure **21** is formed of, for example, oxide, nitride, a high-k dielectric material such as aluminum oxide (AlO), tantalum oxide (TaO), hafnium oxide (HfO), hafnium silicon oxide (HfSiO), hafnium aluminum oxide (HfAO), hafnium tantalum oxide (HfTaO), or a combination thereof.

[0035] The isolation structures **21** divide the substrate **10** into multiple pixel regions **20**. In some embodiments, the isolation structures **21** and the pixel regions **20** are alternately arranged along a second direction **D2** perpendicular to the first direction **D1**. As such, the isolation structure **21** may be referred to as an inter-pixel DTI structure. The isolation structure **21** extends along a third direction **D3** (i.e., into the paper of FIG. 2) perpendicular to the first direction **D1** and the second direction **D2**.

[0036] In some embodiments, the pixel region **20** is formed by providing doping impurities to the substrate **10**, wherein the doping impurities have a conductivity type opposite to a conductivity type of the substrate **10**. In some embodiments, the pixel region **20** is doped with n-type impurities.

Due to their opposite types of dopants, a P-N junction is formed between the substrate **10** and the pixel region **20**. Therefore, each pixel region **20** includes a P-N junction. The P-N junction includes a photodiode. In some embodiments, the pixel region **20** is a light sensing layer. The isolation structure **21** is used to optically isolate the pixel region **20** from an adjacent pixel region **20**.

[0037] Several isolation members **23** are disposed within the substrate **10**. The isolation member **23** is formed of a material different from or same as a material of the isolation structure **21**. In some embodiments, the isolation member **23** extends along the first direction **D1** between the first surface **S1** and the second surface **S2** of the substrate **10**. In some embodiments, the isolation member **23** extends from the second surface **S2** of the substrate **10** toward a predetermined depth of the substrate **10**. In some embodiments, a length **L1** of the isolation member **23** is substantially less than a length **L2** of the isolation structure **21**, as shown in FIG. 2. In other embodiments, the length **L1** of the isolation member **23** is substantially equal to or greater than the length **L2** of the isolation structure **21**. In some embodiments, the isolation member **23** is parallel to the isolation structure **21** along the third direction **D3**.

[0038] In some embodiments, the isolation member **23** is disposed within each pixel region **20** and surrounded by the isolation structures **21**. The isolation member **23** divides the pixel region **20** into at least two photodiode regions **22**. Two photodiode regions **22** in one pixel region **20** are referred to as a dual photodiode (DPD). As such, the isolation member **23** is referred to as an in-pixel DTI structure.

[0039] The photodiode region **22** is used to convert radiation that enters into the substrate **10** from the second surface **S2** of the substrate **10** to an electrical signal. When incident light (containing photons of sufficient energy) strikes the photodiode region **22**, an electron-hole pair is created.

[0040] In some embodiments, the optical portion **P1** includes an anti-reflection layer **24**, multiple metal grids **25**, one or more dielectric layers **26**, multiple color filters **27** and multiple microlenses **28**. The anti-reflection layer **24** is disposed on the second surface **S2** of the substrate **10**. In some embodiments, the anti-reflection layer **24** is formed of oxide, nitride, a high-k dielectric material such as aluminum oxide (AlO), tantalum oxide (TaO), hafnium oxide (HfO), hafnium silicon oxide (HfSiO), hafnium aluminum oxide (HfAlO), hafnium tantalum oxide (HfTaO), or a combination thereof. The anti-reflection layer **24** is used to minimize light reflection and thus allow more light to reach the pixel portion **P2**.

[0041] In some embodiments, the metal grids **25** are disposed on the anti-reflection layer **24** and are aligned with the isolation structures **21**, respectively. The metal grids **25** are formed of tungsten (W), copper (Cu) or aluminum copper (AlCu). The metal grids **25** can be used to reduce optical interference of one pixel region **20** from an adjacent pixel region **20**. The metal grids **25** are used to reflect the refracted light or reflected light back to the color filters **27**, and thus the optical isolation among adjacent pixels can be improved.

[0042] The color filters **27** are disposed on the anti-reflection layer **24** and are near the metal grids **25**. Adjacent pairs of color filters **27** are separated by one metal grid **25**. Space over the metal grids **25** is filled with the dielectric layer **26** made of oxide. The color filters **27** are separated from the substrate **10** or the pixel region **20** by the anti-reflection layer **24**.

[0043] In other embodiments, the dielectric layer **26** is disposed on the anti-reflection layer **24**. In such embodiments, the metal grids **25** are separately embedded in the dielectric layer **26** and are aligned with the isolation structures **21**. That is, the dielectric layer **26** separates the metal grids **25** from the substrate **10**. The color filters **27** are surrounded by the dielectric layer **26**, and top surfaces of the color filters **27** are coplanar with or below a top surface of the dielectric layer **26**.

[0044] In some embodiments, the color filter **27** is aligned with the pixel region **20**. The color filter **27** is disposed over two photodiode regions **22** and an isolation member **23** between the two photodiode regions **22**. Such arrangement can increase radiation of incident light onto the pixel region **20**. The color filter **27** is used to allow light or radiation having a wavelength within a specific range to pass. For example, a color filter **27** used to transmit incident light with a

wavelength between about 620 nanometers (nm) and about 750 nm (i.e., red light) is referred to as a red filter **27R**. A color filter **27** used to transmit incident light with a wavelength between about 495 nm and about 570 nm (i.e., green light) is referred to as a green filter **27G**. A color filter **27** used to transmit incident light with a wavelength between about 450 nm and about 495 nm (i.e., blue light) is referred to as a blue filter **27B**. The red, green and blue filters **27R**, **27G** and **27B** are illustrated in the figures and described below.

[0045] In some embodiments, the microlens **28** is disposed on the color filter **27** and portions of the dielectric layer **26**. The microlens **28** has a curved surface (or convex surface) that directs an incoming light beam and facilitates condensation of the light beam. The microlens **28** is aligned with the color filter **27** and the pixel region **20**. Such arrangement can increase the radiation of the light beam onto the pixel region **20**. Since the photodiode regions **22** are formed as a pair in a single pixel region **20**, one color filter **27** is disposed over two photodiode regions **22**, and one microlens **28** is disposed over two photodiode regions **22**.

[0046] In some embodiments, the circuitry portion **P3** includes one or more transistors **T10**, one or more interlayer dielectric (ILD) layers **42** and multiple conductive lines **44**. The transistor **T10** is disposed on the first surface **S1** of the substrate **10** and is surrounded by the ILD layer **42**. The conductive lines **44** are interconnected and embedded in the ILD layer **42**. The ILD layer **42** is formed of oxide or a suitable material. The conductive lines **44** are formed of a metal or an alloy.

[0047] Although not specifically illustrated in FIG. **2**, the transistor **T10** includes a transfer transistor serving as a transfer gate, a reset transistor serving as a reset gate, a source follower transistor serving as a source follower gate (an amplification gate), and a selection transistor serving as a selection gate. Some or all of the transistors are disposed in the circuitry portion **P3**. In some embodiments, a floating node (or a floating diffusion region) **14** is disposed in the pixel portion **P2**.

[0048] In some embodiments, the floating node **14** is used to store the charges that are transferred and generated from the pixel regions **20**. The charges stored in the floating node **14** are then converted into a voltage signal. Multiple pixel regions **20** can measure different components of the light beam based on multiple voltage signals converted by the floating node **14**. The voltage signals can be read or processed by the circuitry portion **P3** of the image sensing structure **100**. Therefore, a measurement result for generation of a 2D or 3D image of a scene can be provided by the image sensing structure **100**.

[0049] As illustrated in FIG. **1**, all of the isolation members **23** extend along the third direction **D3**. In some embodiments, the isolation members **23** are perpendicular to some of the isolation structures **21**. The CFA **127A** can separate various colors from a color image. A resulting output from the image sensing structure **100** with the CFA **127A** can be interpolated to form a full-color image. In some embodiments, the CFA **127A** is an arrangement of red filters **27R**, green filters **27G** and blue filters **27B** formed in a square over image sensing structures (i.e., the photodiode regions **22**). The red, green and blue filters **27R**, **27G** and **27B** respectively allow incident light with a wavelength within a specific range to pass. The pixel region **20** corresponding to the red filter **27R** is a red pixel region **20R** (or, simply, a red pixel), the pixel region **20** corresponding to the green filter **27G** is a green pixel region **20G** (or, simply, a green pixel), and the pixel region **20** corresponding to the blue filter **27B** is a blue pixel region **20B** (or, simply, a blue pixel). Every four red pixel regions **20R** form a 2×2 red pixel unit **120R**, every four green pixel regions **20G** form a 2×2 green pixel unit **120G**, and every four blue pixel regions **20B** form a 2×2 blue pixel unit **120B**.

[0050] FIG. **3** is a schematic cross-sectional view of a portion of the image sensing structure **100** along a line **BB'** in FIG. **1**. FIG. **3** is used to illustrate an operational principle of the image sensing structure **100**. In some embodiments, a light beam $h\nu$ is incident on the second surface **S2** of the substrate **10** via, in sequence, the microlens **28**, the red filter **27R**/the green filter **27G**, and the anti-reflection layer **24**. A first light beam $h\nu_1$ (i.e. red light in the light beam $h\nu$) can pass through the red filter **27R**, and a second light beam $h\nu_2$ (i.e. green light in the light beam $h\nu$) can pass through

the green filter 27G. After passing through the red filter 27R, the first light hv1 beam splits into at least two red light beams hv1a and hv1b. After passing through the green filter 27G, the second light beam hv2 splits into at least two green light beams hv2a and hv2b. The photodiode regions 22 in FIG. 2 corresponding to the red filter 27R, the green filter 27G and the blue filter 27B are respectively denoted as photodiode regions 22R, photodiode regions 22G, and photodiode regions 22B. The red light beams hv1a and hv1b then interact with the photodiode regions 22R. The green light beams hv2a and hv2b then interact with the photodiode regions 22G. Thus, multiple photo-induced carriers (e.g., electrons) are generated and collected by the photodiode regions 22R and 22G, and the photo-induced carriers are then transferred from the pixel regions 20R and 20G to the floating node 14.

[0051] The circuitry portion P3 processes signals generated from the photo-induced carriers in the pixel portion P2, and performs any suitable operations based on such signals.

[0052] The isolation structures 21 are used to reflect the refracted light or reflected light back to the pixel regions 20, and thus a full-well capacity of the image sensing structure 100 can be increased and optical isolation of adjacent pixels is thus improved.

[0053] However, in some cases, the red light beams hv1a and hv1b may not travel straight along the substrate 10 from the second surface S2 toward the first surface S1 due to their diffraction behavior. Light with longer wavelengths has a greater tendency to diffract or penetrate an obstacle. Accordingly, light with longer wavelengths can reach greater depths of the substrate 10. The red light beams hv1a and hv1b tend to penetrate the isolation structures 21 and reach other adjacent pixel regions 20. Therefore, despite the use of the isolation structures 21, some red light (such as the red light beams hv1a and hv1b) illuminates, for example, adjacent photodiode regions 22G. Such phenomenon is referred to as “crosstalk”. The red light beams hv1a and hv1b penetrating the isolation structures 21 cause some of the photodiode regions 22G illuminated by such red light beams hv1a and hv1b to produce more photo-induced carriers.

[0054] Referring to FIGS. 1 and 3, the red light beams hv1a and hv1b (represented by arrows) penetrating respective neighboring photodiode regions 22G cause some green pixel units 120G to be brighter. The other green pixel units 120G not affected by any red light (i.e., the green pixel units 120G not pointed to by any arrow) is relatively darker. Such problem causes uneven brightness of pixels of the image sensing structure 100.

[0055] FIG. 4 is a schematic top view showing a CFA 127B of another image sensing structure 110. The image sensing structure 110 is similar to the image sensing structure 100, except the image sensing structure 110 has different orientations of isolation members 23. In some embodiments, the image sensing structure 110 includes multiple first isolation members 23X extending along the second direction D2 and multiple second isolation members 23Y extending along the third direction D3. The first isolation member 23X is perpendicular to the second isolation member 23Y from the top view.

[0056] FIGS. 5A and 5B are enlarged top views of portions R10 and R20 of FIG. 4, respectively. Some elements are not shown for clarity. Referring to FIG. 5A, the portion R10 is a red pixel unit 120R including two red pixel regions 20R and two green pixel regions separated by two isolation structures 21 perpendicular to each other. In some embodiments, the red pixel unit 120R includes four second isolation members 23Y extending along the third direction D3 and no first isolation member 23X extending along the second direction D2 is present. The second isolation member 23Y separates two photodiode regions 22R in each red pixel region 20R. The second isolation member 23Y is between the two photodiode regions 22R in each red pixel region 20R. In some embodiments, the second isolation member 23Y is connected to the isolation structure 21.

[0057] Referring to FIG. 5B, the portion R20 includes two red pixel regions 20R and two green pixel regions 20G separated by two isolation structures 21 perpendicular to each other. In some embodiments, the portion R20 includes two first isolation members 23X extending along the second direction D2. The first isolation member 23X separates two photodiode regions 22R in each

red pixel region **20R**. The first isolation member **23X** is between the two photodiode regions **22R** in each red pixel region **20R**. In some embodiments, the first isolation member **23X** is connected to the isolation structure **21**. In some embodiments, the portion **R20** includes two second isolation members **23Y** extending along the third direction **D3**. The second isolation member **23Y** separates two photodiode regions **22G** in each green pixel region **20G**. The second isolation member **23Y** is between the two photodiode regions **22G** in each green pixel region **20G**. In some embodiments, the second isolation member **23Y** is connected to the isolation structure **21**. In some embodiments, a thickness **T1** of the isolation structure **21** is substantially equal to or greater than a thickness **T2** of the first isolation member **23X** or a thickness **T3** of the second isolation member **23Y**. In some embodiments, the thickness **T2** of the first isolation member **23X** is substantially equal to the thickness **T3** of the second isolation member **23Y**.

[0058] FIG. **6A** is a schematic cross-sectional view of the portion **R10** along a line **CC'** in FIG. **4**. FIG. **6B** is a schematic perspective view of the portion **R10** in FIG. **6A**. The isolation structures **21**, the first isolation members **23X** and the second isolation members **23Y** are embedded in the substrate **10**. The isolation structures **21**, the first isolation members **23X** and the second isolation members **23Y** are strips of walls, as shown in FIG. **6B**. For convenience of illustration, only portions of the isolation structures **21** are shown in the perspective views referred to below.

[0059] Similar to the principle discussed above in reference to FIG. **3**, when an incident light beam **hv10** passes through the microlens **28**, the red filter **27R** and the anti-reflection layer **24**, the incident light beam **hv10** is split into at least two red light beams **hv11** and **hv12** by the second isolation member **23Y**. The red light beams **hv11** and **hv12** may penetrate neighboring photodiode regions **22G** in the green pixel region **20G** on opposite sides of the isolation structure **21**.

Therefore, portions of the green pixel region **20G** accepting the additional red light beams (such as **hv11** and **hv12**) will appear brighter. Referring to FIG. **4**, a portion of the green pixel unit **120G** at the right of the portion **R10** (near the arrow) will appear brighter than the rest portion of the green pixel unit **120G**.

[0060] FIG. **7A** is a schematic cross-sectional view of the portion **R20** along a line **DD'** in FIG. **4**. FIG. **7B** is a schematic perspective view of the portion **R20** in FIG. **7A**. Similar to the principle discussed above in reference to FIG. **3**, when an incident light beam **hv20** passes through the microlens **28**, the red filter **27R** and the anti-reflection layer **24**, the incident light beam **hv20** is split into at least two red light beams **hv21** and **hv22** by the first isolation member **23X**. The red light beams **hv21** and **hv22** may penetrate neighboring photodiode regions **22G** in the green pixel region **20G** on opposite sides of the isolation structure **21**. Therefore, portions of the green pixel region **20G** accepting the additional red light beams (such as **hv21** and **hv22**) will appear brighter. Referring to FIG. **4**, a portion of the green pixel unit **120G** at the bottom of the portion **R20** (near the arrow) will appear brighter than the rest portion of the green pixel unit **120G**.

[0061] The image sensing structure **110** includes both the first isolation members **23X** and the second isolation members **23Y**. Compared with the isolation member **23Y** in the portion **R10**, the first isolation member **23X** in the portion **R20** has been “rotated” 90 degrees. As such, in each of the green pixel units **120G**, two green pixel regions **20G** accept the red light beams coming from neighboring red pixel regions **20R**, as shown by the arrows in FIG. **4**. Therefore, all the green pixel units **120G** have substantially a same average brightness (i.e., half bright and half dark). A brightness uniformity of pixels of the image sensing structure **110** is thus improved. In some embodiments, the image sensing structure **110** has a better color performance when all the green pixel units **120G** have the same average brightness compared with a design that some of the green pixel units **120G** are physically bright and the other of the green pixel units **120G** are physically dark.

[0062] FIGS. **8** to **10** are schematic perspective views of various portions **R30**, **R40** and **R50** of FIG. **4**. Referring to FIGS. **4** and **8**, the portion **R30** includes one red pixel region **20R**, two green pixel regions **20G** and one blue pixel region **20B** adjacent to each other and separated by two

isolation structures **21** perpendicular to each other. The portion **R30** includes four second isolation members **23Y** extending along the second direction **D2** and no first isolation member **23X** extending along the first direction **D1**. The second isolation members **23Y** are disposed in the red pixel regions **20R**, **20G** and **20B**.

[0063] Referring to FIGS. **4** and **9**, the portion **R40** includes one red pixel unit **120R** and one green pixel unit **120G** adjacent to each other and separated by one isolation structure **21**. The portion **R40** includes four isolation member **23X** extending along the first direction **D1** and four isolation member **23Y** extending along the second direction **D2**.

[0064] Referring to FIGS. **4** and **10**, the portion **R50** includes one red pixel unit **120R** and one green pixel unit **120G** adjacent to each other and separated by one isolation structure **21**. The portion **R50** is similar to the portion **R40**, except that in the portion **R40**, the first isolation members **23X** are disposed in the red pixel unit **120R** and the second isolation members **23Y** are disposed in the green pixel unit **120G**, while in the portion **R50**, the first isolation members **23X** are disposed in the green pixel unit **120G** and the second isolation members **23Y** are disposed in the red pixel unit **120R**.

[0065] FIG. **11** is a schematic top view showing a CFA **127C** of another image sensing structure **120**. The image sensing structure **120** is similar to the image sensing structure **100** or **110**, except the image sensing structure **120** has different orientations of isolation members. In the image sensing structure **120**, all the green pixel units **120G** have substantially a same average brightness because two green pixel regions **20G** in each green pixel unit **120G** accept red light beams coming from neighboring red pixel regions **20R**, as shown in FIG. **11** by arrows. A brightness uniformity of pixels of the image sensing structure **120** is thus improved.

[0066] FIG. **12A** is a schematic cross-sectional view of a portion **R60** along a line **EE'** in FIG. **11**. FIG. **12B** is a schematic perspective view of the portion **R60** in FIG. **12A**. The portion **R60** is a blue pixel unit **120B** including four blue pixel regions **20B** adjacent to each other and separated by two isolation structures **21** perpendicular to each other. In some embodiments, the portion **R60** includes four second isolation members **23Y** extending along the second direction **D2**. Each isolation member **23Y** is disposed between two photodiode regions **22B**, as shown in FIG. **12A** or **12B**. The isolation member **23Y** separates two photodiode regions **22B** within the blue pixel region **20B**. Therefore, there are eight photodiode regions **22B** in the portion **R60** (the blue pixel unit **120B**).

[0067] FIG. **13A** is a schematic cross-sectional view of a portion **R70** along a line **FF'** in FIG. **11**. FIG. **13B** is a schematic perspective view of the portion **R70** in FIG. **13A**. The portion **R70** is another blue pixel unit **120B** similar to the portion **R60**, except the portion **R70** includes four first isolation members **23X** extending along the first direction **D1**. Each first isolation member **23X** is disposed between two photodiode regions **22B** within the blue pixel region **20B**. Compared to the second isolation members **23Y** in the portion **R60**, the first isolation members **23X** in the portion **R70** have been “rotated” 90 degrees. Therefore, not only can the isolation member in the red pixel unit **120R** be rotated, but the isolation member in the blue pixel unit **120B** can also be rotated.

[0068] FIGS. **14** and **15** are schematic perspective views of portions **R80** and **R90** of FIG. **11**. Referring to FIGS. **11** and **14**, the portion **R80** includes one red pixel region **20R**, two green pixel regions **20G** and one blue pixel region **20B** adjacent to each other and separated by two isolation structures **21** perpendicular to each other. The portion **R80** includes one first isolation member **23X** extending along the first direction **D1** and three second isolation members **23Y** extending along the second direction **D2**. The first isolation member **23X** is disposed in the red pixel region **20R**, and the second isolation members **23Y** are disposed in the green pixel regions **20G** and the blue pixel region **20B**.

[0069] Referring to FIGS. **11** and **15**, the portion **R90** includes one red pixel region **20R**, two green pixel regions **20G** and one blue pixel region **20B** adjacent to each other and separated by two isolation structures **21** perpendicular to each other. The portion **R90** includes one first isolation

member **23X** extending along the first direction **D1** and three second isolation members **23Y** extending along the second direction **D2**. The first isolation member **23X** is disposed in the blue pixel region **20B**, and the second isolation members **23Y** are disposed in the red pixel region **20R** and the green pixel regions **20G**.

[0070] FIG. **16** is a schematic top view showing a CFA **127D** of another image sensing structure **130**. The image sensing structure **130** is similar to the image sensing structure **100**, **110** or **120**, except the image sensing structure **130** has different orientations of isolation members. In some embodiments, the image sensing structure **130** includes a portion **R100** and a portion **R110**, each of which includes one red pixel unit **120R**, two green pixel units **120G** and one blue pixel unit **120B** separated by the isolation structure **21**. The portion **R100** includes multiple second isolation members **23Y** extending along the second direction **D2** and no first isolation member **23X**, while the portion **R110** includes multiple first isolation members **23X** extending along the first direction **D1** and no isolation member **23Y**. Compared to the second isolation members **23Y** in the portion **R100**, the first isolation members **23X** in the portion **R110** have been “rotated” 90 degrees. In the image sensing structure **130**, all the green pixel units **120G** have substantially a same average brightness because two green pixel regions **20G** in each green pixel unit **120G** accept red light beams coming from neighboring red pixel regions **20R**, as shown in FIG. **16** by arrows. A brightness uniformity of pixels of the image sensing structure **130** is thus improved.

[0071] FIG. **17** is a schematic top view showing a CFA **127E** of another image sensing structure **140**. The image sensing structure **140** is similar to the image sensing structure **100**, **110**, **120** or **130** except the image sensing structure **140** has different orientations of isolation members. The image sensing structure **140** includes multiple red pixel units **120R**, multiple green pixel units **120G** and multiple blue pixel units **120B**. In some embodiments, multiple first isolation members **23X** extending along the first direction **D1** and multiple second isolation members **23Y** extending along the second direction **D2** are disposed in each of the red pixel units **120R**, the green pixel units **120G** and the blue pixel units **120B**. The first isolation members **23X** and the second isolation members **23Y** may be alternately arranged. In some embodiments, each of the red pixel units **120R**, the green pixel units **120G** and the blue pixel units **120B** includes 50% first isolation members **23X** and 50% second isolation members **23Y**. That is, half of the isolation members in each pixel unit are rotated 90 degrees, and another half of the isolation members are not rotated.

[0072] FIG. **18A** is an enlarged view of a portion **R120** of the image sensing structure **140** of FIG. **17**. FIGS. **18B** and **18C** are schematic cross-sectional and perspective views of the portion **R120** of FIG. **18A**. Referring to FIG. **18A**, the portion **R120** is a red pixel unit **120R** including four red pixel regions **20R** adjacent to each other and separated by two isolation structures **21** perpendicular to each other. In some embodiments, the red pixel unit **120R** includes two first isolation members **23X** extending along the first direction **D1** and two second isolation members **23Y** extending along the second direction **D2**. In some embodiments, the two first isolation members **23X** are connected to a same isolation structure **21**, and the two second isolation members **23Y** are connected to another isolation structure **21**.

[0073] From the top view of FIG. **18A**, the two first isolation members **23X** are positioned diagonally opposite to each other, and the two second isolation members **23Y** are positioned diagonally opposite to each other. As seen in FIG. **18A**, in the upper left red pixel region **20R** or the lower right red pixel region **20R**, the two photodiode regions **22R** are separated to left and right by the in-pixel DTI **23Y**. That is, the two photodiode regions **22R** are arranged along the first direction **D1**. As seen in FIG. **18A**, in the upper right red pixel region **20R** or the lower left red pixel region **20R**, the two photodiode regions **22R** are separated to front and back by the first isolation member **23X**. That is, the pair of photodiode regions **22R** are arranged along the second direction **D2**. FIGS. **18B** and **18C** illustrate the first isolation members **23X**, the second isolation members **23Y** and the isolation structures **21** embedded in the substrate **10** and extending along specific directions in the substrate **10**.

[0074] FIGS. **19A** and **19B** are schematic cross-sectional and perspective views of the portion **R130** of the image sensing structure **140** of FIG. **17**. Referring to FIGS. **17**, **19A** and **19B**, the portion **R130** includes one red pixel region **20R**, two green pixel regions **20G** and one blue pixel region **20B** adjacent to each other and separated by two isolation structures **21** perpendicular to each other. The portion **R130** includes one first isolation member **23X** in the red pixel region **20R**, one first isolation member **23X** in the blue pixel region **20B** and two first isolation members **23X** respectively disposed in the green pixel regions **20G**. The first isolation members **23X**, the second isolation members **23Y** and the isolation structures **21** are embedded in the substrate **10** and extend along specific directions in the substrate **10**.

[0075] In the image sensing structure **140**, all the green pixel units **120G** have substantially a same average brightness because two green pixel regions **20G** in each green pixel unit **120G** accept red light beams coming from neighboring red pixel regions **20R**, as shown in FIG. **17** by arrows. A brightness uniformity of pixels of the image sensing structure **140** is thus improved.

[0076] FIG. **20** is a flow diagram showing a method **200** for forming an image sensing structure **110**. FIGS. **21** to **27** are schematic cross-sectional or top views illustrating sequential operations of the method **200** in FIG. **20**. Operations **201**, **203**, **205**, **207**, **209** and **211** are used to form a pixel portion **P2** connected to a circuitry portion **P3**.

[0077] For forming the pixel portion **P2**, in operation **201** of FIG. **20**, a substrate **10** is provided, as shown in FIG. **21**. The substrate **10** has a first surface **S1** (or a front side **S1**) and a second surface **S2** (or a back side **S2**) opposite to the first surface **S1**. The substrate **10** includes any type of semiconductor body such as a silicon (Si) wafer, a silicon-germanium (SiGe) wafer, a silicon-on-insulator (SOI) substrate, or the like. Although not shown, the substrate **10** includes one or more semiconductor layers and/or epitaxial layers formed thereon. In some embodiments, the substrate **10** is doped with p-type impurities and thus forms a p-type substrate **10**.

[0078] In operation **203** of FIG. **20**, a transistor **T10** is formed on the first surface **S1** of the substrate **10**, as shown in FIG. **22**. The transistor **T10** includes a gate structure and a source/drain structure, which may be formed using suitable methods such as photolithography, etching, epitaxy, implantation or deposition. In some embodiments, the transistor **T10** includes a transfer transistor serving as a transfer gate, a reset transistor serving as a reset gate, a source follower transistor serving as a source follower gate (an amplification gate), and a selection transistor serving as a selection gate. In some embodiments, a floating node (or a floating diffusion region) **14** is formed within the substrate **10** and proximal to the first surface **S1**. The floating node **14** is formed as a high implant (e.g., N⁺ implant) in the substrate **10**. The floating node **14** is electrically connected to the source follower transistor and/or the selection transistor.

[0079] In operation **205** of FIG. **20**, the circuitry portion **P3** (or an interconnect portion **P3**) is formed on the first surface **S1** of the substrate **10**, as shown in FIG. **23**. The circuitry portion **P3** includes one or more interlayer dielectric (ILD) layers **42** and multiple conductive lines **44**. The circuitry portion **P3** can be formed using suitable methods such as photolithography, etching, deposition, electroplating or planarization. The ILD layer **42** may be formed of oxide or another suitable material. The ILD layer **42** may be formed of one or more of a low-k dielectric layer (i.e., a dielectric material with a dielectric constant less than about 3.9), an ultra low-k dielectric layer, or an oxide (e.g., silicon oxide). The conductive lines **44** can be formed of copper (Cu), cobalt (Co), aluminum (Al), silver (Ag), gold (Au), tungsten (W), the like, or a combination thereof. The conductive lines **44** are interconnected by conductive vias (not shown) and embedded in the ILD layer **42**. The transistor **T10** is covered and surrounded by the ILD layer **42**. The circuitry portion **P3** is electrically connected to the transistor **T10**.

[0080] In operation **207** of FIG. **20**, multiple isolation structures **21** are formed in the substrate **10**, as shown in FIGS. **24A** and **24B**. Referring to FIG. **24A**, the substrate **10** is flipped with the second surface **S2** facing upward. Referring to FIG. **24B**, although not specifically illustrated, photolithographic or etching operations are used to form multiple trenches extending from the

second surface **S2** of the substrate **10** to a predetermined depth of the substrate **10**. A dielectric material is used to fill the trenches using a deposition operation. The dielectric material includes, for example, silicon oxide (SiO), silicon nitride (SiN), aluminum oxide (AlO), tantalum oxide (TaO), hafnium oxide (HfO), hafnium silicon oxide (HfSiO), hafnium aluminum oxide (HfAlO), hafnium tantalum oxide (HfTaO), or a combination thereof.

[0081] In operation **209** of FIG. **20**, multiple pixel regions **20** are formed within the substrate **10**, as shown in FIGS. **25A** and **25B**. Referring to FIG. **25A**, an implantation operation is used to dope n-type impurities **18** into the substrate **10**. Referring to FIG. **25B**, the substrate **10** doped with the n-type impurities **18** to form the pixel regions **20**. The pixel region **20** includes a P-N junction formed between the un-doped substrate **10** (which is p-type) and the doped substrate **10** (which is n-type). The P-N junction may form a photodiode. The pixel regions **20** are separated by the isolation structures **21**.

[0082] In operation **211** of FIG. **20**, multiple first isolation members **23X** and second isolation members **23Y** are formed in the substrate **10**, as shown in FIGS. **26A** to **26E**. Referring to FIG. **26A**, a photoresist pattern **40** is formed on the substrate **10** using a photomask. The photoresist pattern **40** includes multiple openings **O1** exposing portions of the second surface **S2** corresponding to the pixel regions **20**.

[0083] Referring to FIG. **26B**, FIG. **26B** is a schematic top view of FIG. **26A**. In some embodiments, the openings **O1** have a strip shape. In some embodiments, the openings **O1** include first openings **O1X** extending along the second direction **D2** and second openings **O1Y** extending along the third direction **D3**. In some embodiments, the first opening **O1X** is substantially orthogonal to the second opening **O1Y**.

[0084] Referring to FIG. **26C**, an etching operation is used to remove portions of the substrate **10** using the photoresist pattern **40** as an etching mask. The photoresist pattern **40** is then removed. As a result, multiple trenches **O2** are formed in the substrate **10**. The trenches **O2** extend from the second surface **S2** of the substrate **10** to a predetermined depth of the substrate **10**.

[0085] Referring to FIG. **26D**, a deposition operation is used to deposit a dielectric material into the trenches **O2**. The dielectric material includes, for example, silicon oxide (SiO), silicon nitride (SiN), aluminum oxide (AlO), tantalum oxide (TaO), hafnium oxide (HfO), hafnium silicon oxide (HfSiO), hafnium aluminum oxide (HfAlO), hafnium tantalum oxide (HaTaO), or a combination thereof. A planarization operation may be used to remove excess dielectric material. Therefore, the isolation members **23** are formed in the substrate **10**. The isolation member **23** has a top surface coplanar with the second surface **S2** of the substrate **10** or the isolation structure **21**.

[0086] Referring to FIG. **26E**, FIG. **26E** is a schematic top view of the structure of FIG. **26D**. In some embodiments, the isolation members **23** have a strip shape. In some embodiments, the isolation members **23** include first isolation members **23X** extending along the second direction **D2** and second isolation members **23Y** extending along the third direction **D3**. In some embodiments, the first isolation members **23X** and the second isolation members **23Y** are connected to the isolation structures **21**. Therefore, formation of the pixel portion **P2** and the circuitry portion **P3** is complete.

[0087] In operation **213** of FIG. **20**, an optical portion **P1** is formed on the second surface **S2** of the substrate **10**, as shown in FIG. **27**. In some embodiments, the optical portion **P1** includes an anti-reflection layer **24**, multiple metal grids **25**, one or more dielectric layers **26**, multiple color filters **27**, and multiple microlenses **28**. The anti-reflection layer **24** is formed on the second surface **S2** of the substrate **10**. In some embodiments, the anti-reflection layer **24** is formed of oxide, nitride, a high-k dielectric material such as aluminum oxide (AlO), tantalum oxide (TaO), hafnium oxide (HfO), hafnium silicon oxide (HfSiO), hafnium aluminum oxide (HfAlO), hafnium tantalum oxide (HfTaO), or a combination thereof. The anti-reflection layer **24** may minimize light reflection and thus allow more light to reach the pixel regions **20**.

[0088] In some embodiments, the metal grids **25** are formed on the anti-reflection layer **24** and

aligned with the isolation structures **21**, respectively. The metal grids **25** may be formed of tungsten (W), copper (Cu) or aluminum copper (AlCu). The metal grids **25** can be used to reduce optical interference of one pixel region **20** from an adjacent pixel region **20**.

[0089] The color filters **27** are formed on the anti-reflection layer **24** and near to the metal grids **25**. Adjacent color filters **27** are separated by one metal grid **25**. Space over the metal grids **25** may be filled with the dielectric layer **26** made of oxide. The color filters **27** are separated from the substrate **10** or the pixel region **20** by the anti-reflection layer **24**.

[0090] In some embodiments, each color filter **27** is aligned with a corresponding pixel region **20**. The color filter **27** is formed over two photodiode regions **22** and an isolation member **23** between the two photodiode regions **22**. Such arrangement can increase radiation of incident light onto the pixel region **20**. The color filter **27** is used to allow light or radiation having wavelengths within a specific range to pass. For example, a color filter **27** used to transmit incident light with a wavelength between about 620 nanometers (nm) and about 750 nm (i.e., red light) is referred to as a red filter **27R**. A color filter **27** used to transmit incident light with a wavelength between about 495 nm and about 570 nm (i.e., green light) is referred to as a green filter **27G**. A color filter **27** used to transmit incident light with a wavelength between about 450 nm and about 495 nm (i.e., blue light) is referred to as a blue filter **27B**.

[0091] In some embodiments, the microlens **28** is formed on the color filter **27** and portions of the dielectric layer **26**. The microlens **28** has a curved surface (or convex surface) that directs an incoming light and facilitates condensation of the incident light. The microlens **28** is aligned with the color filter **27** and the pixel region **20**. Such arrangement can increase radiation of the incident light on the pixel region **20**. Since the photodiode regions **22** are formed in pairs in a single pixel region **20**, a color filter **27** is formed over two photodiode regions **22**, and a microlens **28** is formed over two photodiode regions **22**. As a result, the formation of the image sensing structure **110** is complete.

[0092] FIG. **28** is a schematic perspective view illustrating a combination of two image sensing structures **150** and **160**. In some embodiments, the image sensing structure **150** is a portion of an image sensing device formed on a first wafer, and the image sensing structure **160** is a portion of an image sensing device formed on a second wafer. In some embodiments, the image sensing structure **150** includes multiple first isolation members **23X** and second isolation members **23Y**. For example, the first isolation members **23X** may be disposed in the red pixel unit **120R** and the blue pixel unit **120B**, but the present disclosure is not limited thereto. In some embodiments, the image sensing structure **160** includes multiple second isolation members **23Y**. Compared with the image sensing structure **160**, half of the second isolation members **23Y** in the image sensing structure **150** are rotated 90 degrees to form the first isolation members **23X**.

[0093] In some embodiments, the image sensing structure **150** includes multiple first transistors **T12** symmetrically arranged with each other. The first transistors **T12** include transfer (TX) transistors or transfer gates. In some embodiments, the first transistors **T12** are disposed below and at both sides of one first isolation member **23X**. In some embodiments, the first transistors **T12** are disposed below and at both sides of one second isolation member **23Y**. In some embodiments, the image sensing structure **160** includes multiple second transistors **T14** disposed adjacent to the second isolation members **23Y**. An arrangement of the second transistors **T14** may be parallel to an extension direction of the second isolation member **23Y**. The second transistors **T14** include reset (RST) transistors, source follower (SF) transistors, row selection (SEL) transistors, or the like. In some embodiments, the image sensing structure **150** is aligned with and bonded to the image sensing structure **160**. In such embodiments, the first transistors **T12** may be electrically connected to the second transistors **T14**. With such design, layout of transistors and wirings of image sensing structures can be simplified. In addition, uneven brightness of pixels can also be solved.

[0094] One aspect of the present disclosure provides an image sensing structure. The image sensing structure includes: a first pixel, including a first photodiode, a second photodiode and a first

isolation member disposed between the first photodiode and the second photodiode and extending along a first direction; and a second pixel, disposed adjacent to the first pixel and including a third photodiode, a fourth photodiode and a second isolation member disposed between the third photodiode and the fourth photodiode and extending along a second direction substantially perpendicular to the first direction.

[0095] One aspect of the present disclosure provides another image sensing structure. The image sensing structure includes: a first sensing member including a plurality of first pixels adjacent to and separated from each other, wherein each of the plurality of first pixels includes a first photodiode, a second photodiode and a first isolation member disposed between the first photodiode and the second diode; and a second sensing member, adjacent to the first sensing member and including a plurality of second pixels adjacent to and separated from each other, wherein each of the plurality of second pixels includes a third photodiode, a fourth photodiode and a second isolation member disposed between the third photodiode and the fourth photodiode, wherein at least one of the first isolation members extends along a first direction, and at least one of the second isolation members extends along a second direction substantially perpendicular to the first direction.

[0096] Another aspect of the present disclosure provides a method of forming an image sensing structure. The method includes: providing a substrate having a first surface and a second surface opposite to the first surface; forming a first photodiode and a second photodiode in the substrate; forming a third photodiode and a fourth photodiode in the substrate; forming a first isolation member extending within the substrate and between the first photodiode and the second photodiode; and forming a second isolation member adjacent to the first isolation member and extending within the substrate and between the third photodiode and the fourth photodiode, wherein the first isolation member extends along a first direction, and the second isolation member extends along a second direction substantially perpendicular to the first direction.

[0097] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other operations and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

[0098] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein be utilized according to the present invention. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods and steps.

Claims

1. An image sensing structure, comprising: a first pixel, including a first photodiode, a second photodiode, and a first isolation member, the first isolation member disposed between the first photodiode and the second photodiode and extending along a first direction; and a second pixel, disposed adjacent to the first pixel and including a third photodiode, a fourth photodiode, and a second isolation member, the second isolation member disposed between the third photodiode and

the fourth photodiode and extending along a second direction substantially perpendicular to the first direction.

2. The image sensing structure of claim 1, wherein the first pixel includes a first color filter disposed over the first photodiode, the second photodiode, and the first isolation member, and the second pixel includes a second color filter disposed over the third photodiode, the fourth photodiode, and the second isolation member.

3. The image sensing structure of claim 2, wherein the first color filter is a red filter, and the second color filter is a green filter or blue filter.

4. The image sensing structure of claim 2, wherein the first color filter is a red filter or blue filter, and the second color filter is a green filter.

5. The image sensing structure of claim 2, wherein the first color filter and the second color filter are red filters.

6. The image sensing structure of claim 1, further comprising a third isolation member disposed between the first pixel and the second pixel and extending along the second direction.

7. The image sensing structure of claim 6, wherein the third isolation member is disposed between the first photodiode and the third photodiode, and between the first photodiode and the fourth photodiode.

8. The image sensing structure of claim 6, wherein a thickness of the first isolation member and a thickness of the second isolation member are respectively less than a thickness of the third isolation member.

9. The image sensing structure of claim 1, wherein a second image sensing structure is bonded to the image sensing structure, the image sensing structure includes a first transistor, and the second image sensing structure includes a second transistor electrically connected to the first transistor.

10. The image sensing structure of claim 9, wherein the first transistor includes a transfer transistor, and the second transistor includes a reset transistor, a source follower transistor or a row selection transistor.

11. An image sensing structure, comprising: a first sensing member including a plurality of first pixels adjacent to and separated from each other, wherein each of the plurality of first pixels includes a first photodiode, a second photodiode, and a first isolation member, the first isolation member disposed between the first photodiode and the second photodiode; and a second sensing member, adjacent to the first sensing member and including a plurality of second pixels adjacent to and separated from each other, wherein each of the plurality of second pixels includes a third photodiode, a fourth photodiode, and a second isolation member, the second isolation member disposed between the third photodiode and the fourth photodiode, wherein at least one of the first isolation members extends along a first direction, and at least one of the second isolation members extends along a second direction substantially perpendicular to the first direction.

12. The image sensing structure of claim 11, wherein all of the first isolation members extend along the first direction, and all of the second isolation members extend along the second direction.

13. The image sensing structure of claim 11, wherein each of the plurality of first pixels is a red pixel, and each of the plurality of second pixels is a green pixel or a blue pixel.

14. The image sensing structure of claim 11, wherein each of the plurality of first pixels is a red pixel or a blue pixel, and each of the plurality of second pixels is a green pixel.

15. The image sensing structure of claim 11, wherein the plurality of first pixels include a first red pixel, a first green pixel, and a first blue pixel, and the plurality of second pixels include a second red pixel, a second green pixel, and a second blue pixel.

16. A method of forming an image sensing structure, comprising: providing a substrate having a first surface and a second surface opposite to the first surface; forming a first photodiode and a second photodiode in the substrate; forming a third photodiode and a fourth photodiode in the substrate; forming a first isolation member extending within the substrate and between the first photodiode and the second photodiode; and forming a second isolation member adjacent to the first

isolation member and extending within the substrate and between the third photodiode and the fourth photodiode, wherein the first isolation member extends along a first direction, and the second isolation member extends along a second direction substantially perpendicular to the first direction.

17. The method of claim 16, wherein the first photodiode and the second photodiode extend along the first direction, and the third photodiode and the fourth photodiode extend along the second direction, and the first photodiode, the second photodiode, the third photodiode, and the fourth photodiode are formed by ion implantation operation.

18. The method of claim 16, wherein the formation of the first photodiode and the second photodiode and the formation of the third photodiode and the fourth photodiode are performed prior to the formation of the first isolation member and prior to the formation of the second isolation member.

19. The method of claim 16, wherein the formation of the first isolation member includes removing a first portion of the substrate from the first surface toward the second surface to form a first trench and filling the first trench by a dielectric material, the formation of the second isolation member includes removing a second portion of the substrate from the first surface toward the second surface to form a second trench and filling the second trench by the dielectric material.

20. The method of claim 19, wherein the first trench is substantially perpendicular to the second trench.
